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Using Barcode to Track Student Attendance and Assets in Higher Education Institutions

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Abstract

Most instructors and staff in a university environment use manual methods of tracking student attendance and college assets. However, these methods have several disadvantages; the major drawbacks of taking attendance manually are errors in data collection and the loss of part of the lecture time. As for asset tracking, the common issues are human errors in data collection, wrong inventory estimation and time wastage during the process. Automated identification and data capturing technologies can help address these issues. This paper investigates the implementation of one such system, the barcode technology, to read and process student attendance data in classroom settings and to track periodic inventory of assets within the college facilities such as devices, equipment, office furniture, classrooms, and laboratories. The paper measures faculty members and staff response on usability, comfort level, and efficiency for the proposed barcode system. This work shows that barcode technology eliminates data entry errors and improves efficiency for entering data with a minimum cost of creating barcode tags and hardware.

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1. Introduction and literature review

Tracking student attendance and assets are two important problems in most universities since these are parts of the evaluation process for students and annual audits for university assets. While there are many methods in the literature to solve the student and assets tracking, we choose barcode technology because it is cheap and easy to implement. In this paper, we start with a historical overview of the beginning of the use of barcode technology and identifying some of its types. Bernard Silver and Norman Joseph Woodland invented the first optically scanned barcode, introduced the first-ever barcode, which looks like a bullseye. In 1952, they developed the barcode and reduced the cost of the system and registered a patent. Later, they developed solutions to automatically read product information during grocery checkout for the food chain Food Fair. Since then, the barcode project has widely spread. In the 1970s, a Universal Product Code (UPC) was scanned for the first time at a grocery store market in the United States. The automotive industry and US Department of Defense (DoD) adopted another barcode standard, the Code 39 [1].

Nowadays, barcodes are used on most goods and materials. The barcode technology's key features are easy to use, low cost, and its widespread use in various fields including products and services. Barcode technology uses optical scanners to read data such as numbers, letters, images, and locations. It can be a one-dimensional code or two-dimensional code. One-dimensional code is designed in the form of black rectangle bars called barcode with limited features. However, two-dimensional code is designed as boxes, hexagonal shapes, and other geometric shapes with many features. Both dimensional codes are described in Table 1.

Table. 1. Illustrates the main characteristics of one and two-dimensional code.

Features of one-dimensional code	Features of two-dimensional code
One Dimensional (1D) barcodes	Two Dimensional barcodes (QR Code)
Consists of white and black parallel lines of varying widths with specific gaps	Represent more data per unit area than 1D barcodes
Cannot carry much information	Can carry different type of data like images, locations, cost, and quality
Data type that can carry numbers and text	High data density
Used in retail, warehousing, healthcare, and education	Generally, encode a wide variety of data type in square or rectangular patterns
Examples: Universal Product Codes (UPCs) and International Standard Book Number (ISBN)	Used in consumer advertising, code payments, website logins

Automatic Identification and Data Capture (AIDC) represents a group of technologies that support direct data entry into a computer without using a keyboard [2]. AIDC technologies can improve accuracy, efficiency and reduce the time required for identification, marking and data collection [3]. Almost all the AIDC technologies are comprised of three principal components: (1) data encoder, (2) machine reader or scanner, and (3) data decoder [4]. The data encoder converts alphanumeric characters into machine-readable code whereas the machine reader or scanner reads the encoded data and transforms it into an analog electrical signal. Finally, the data decoder converts the electrical signal into digital form and then into an original alphanumeric characters data form. The AIDC technologies have been successfully applied in the manufacturing, retail, and logistics industry [5]. However, their implementation in the education sector is not that common. The authors view this as a great opportunity for the education sector, where these technologies can be applied to automate a wide range of activities, including student attendance and asset tracking.

Key AIDC technologies include barcodes, radio frequency identification (RFID), magnetic stripe, optical character recognition, biometrics and machine vision [6]. Barcodes and RFIDs are the two most common and frequently used AIDC technologies [7][8]. "A barcode, consisting of bars and spaces, is a machine-readable representation of numerals and characters" [9]. RFID is an electromagnetic based technology that uses a wireless tag to transfer and receive data [2]. The tags are generally attached with items to identify and track them automatically.

Various authors have reported the use of AIDC technologies in the education sector. For example, Rădulescu and Popescu [10] explored the use of barcode technology for tracking and managing library books. Singh [11] has proposed the use of near field communication (NFC), a type of RFIDs, for library management. Rios-Aguilar et al. [12] proposed the use of NFC based mobile business information system to register class attendance. Eydgahi [13] has explored the use of RFIDs to manage attendance for undergraduate senior design projects.

Some other researchers have also explored ways of automating students' attendance. For instance, Chaniago and Junaidi [14] proposed automating student attendance using barcode technology and then sending text messages to their parents using their registered mobile phones. Chintalapudi et al., [15] suggested using a machine-learning based face recognition system to record student attendance. Hajri et al., [16] built an android based application embedded with face recognition, biometrics, and RFID system. Nordin and Fauzi [17] proposed a web-based mobile attendance system with an integrated facial recognition feature. Rao, J Bhaskara et al., [18] also worked on an automated attendance system that detects and recognizes student face to record the attendance.

Researchers have also explored the use of AIDC technologies for asset tracking. For instance, Chew et al., [19] presented the use of barcode technology for maintenance management on the production floor. Huddud et al., [20] explored the use of RFIDs for asset tracking in the manufacturing industry. Lugo et al., [21] investigated the use of RFIDs in securing documents and tracking assets in electronics manufacturing. Mark and Farag [22] reported that a gas and electric company was working to build a software solution using barcode technology and GPS for asset tracking and traceability.

The choice of the above-mentioned technologies depends on application area, available budget, existing infrastructure, and management support. The use of barcode technology is comparatively cheaper as compared to other AIDC technologies [23]. This work considers the implementation of barcode technology in a public sector university because it is a cheaper and more viable option for attendance monitoring and asset tracking. Another reason is that the students' Identity Cards (IDs) have already printed barcodes on them, so the additional cost will not be incurred other than the barcode readers. As far as asset tracking is concerned, printed barcodes can be attached to them for tracking purposes. Therefore, we take the view that the barcode system offers a convenient and cost-effective method of tracking students' attendance and assets in a university setting compared to other technologies.

This paper makes two contributions, First, it provides a simple methodology for tracking student attendance based on the printed barcode in student ID, which may later link the information of student attendance to university Edugate. In order to retrieve any information related to student evaluation, registration, and/or preparing reports, the Edugate database can be used as a resource to provide this information to instructors, students and the university administration. Second, asset tracking can be conducted in a short period of time and store the data in electronic mode instead of paperwork. The remainder of the paper is structured as follows: In section 2, we present the tools and hardware used during this research work. The examples and results that represent the conducted experimental work for student attendance and assets tracking are presented in section 3. Finally, the conclusion and direction for future work are presented in section 4.

2. Tools and hardware used

Several software tools and hardware have been used in this research work. Among the tools that we used are barcode generators and barcode scanner application which will be discussed in more details in sections 2.1 and 2.2,

respectively. In this section, we will exhibit the equipment used to read and print the barcode. Reading a barcode requires the code to be printed clearly and read with suitable reading equipment [24].

Fig. 1. (a) Barcode reader (Zebra DS9208); (b) Barcode printer (Zebra GK420T).

The selected barcode reader (Zebra Motorola Symbol DS9208 Handheld 2D) can scan 1-D, 2-D and PDF417 barcodes on paper, mobile phone and computer display and scan virtually any barcode on any medium. Depending on the required barcode quality for this research, we choose a barcode printer (Zebra GK420D direct thermal printer with Ethernet) that can print both direct thermal and thermal transfer labels 2D and QR barcode, and has compatible windows-based software/systems. Figure 1 (a) and (b) illustrate the used barcode reader and the barcode printer, respectively. The other used hardware is a laptop (Microsoft surface) and a smartphone (Samsung Galaxy Note 8).

2.1 Barcode Generation

In this section, steps to create a cost-effective barcode system are discussed. Section 3.2 explains how barcode labels can be attached to a selected asset at the college meetings room. The barcode is created using an open-source program application that can be downloaded through a computer or cell phone, including the program (QR create). This program allows creating a barcode for websites, texts, locations, and other services. The benefits of creating our own barcodes are the free and fast process. However, it works only for a specific operation. Excel spreadsheet is used to facilitate and present our data. In the beginning, we identified some open source programs application to create a barcode. Examples of these programs include Zint Barcode Generator, 2D Barcode Generator, Barillo Barcode, Bytescout Barcode Generator, Dataware Barcode, Labeljoy 6 Barcode. Most of these softwares were downloaded during our research work to see how they work. After some attempts to try these applications and software tools, the Zint Barcode Generator is selected to create the barcodes. Zint Barcode is chosen because the program provides open-source implementation for encoding data in a range of barcode symbologists. Barcode Active program is added to the Microsoft Excel program as an add-in to create barcodes that allows creating a large number of barcodes faster [25].

2.2 Barcode Scanner Applications

In this section, we will discuss how barcode scanner applications can be used to read barcode labels. Several barcode scanner applications were downloaded to see how they can be used. During this process, various scanning programs were explored, such as QR Scanning, Barcode Reader, and Scan-IT to Office. After some attempts and observation of how these apps and software tools work, the application “Scan- IT o Office Program” is selected and considered to be used in our research work because it allows transferring data and information from phone to Microsoft office program.

For Scan-IT to Office. There are a set of steps to install the program and use it along with excel spreadsheet. First, download it to Excel by inserting then add-ins. Second, download Scan-IT to Office application in the cell phone. Third, connect the cell phone and computer by Wi-Fi. Third, click add-ins and click the program. Fourth, a barcode

will appear on the screen. Fifth, scan the barcode by Scan-IT to Office- on phone, and thus the program is connected to Excel spreadsheet.

3. Examples and results

In this section, we will discuss the experimental works conducted during the research work. First, we will explain how barcode can be used for student attendance in a real classroom within the college of engineering. Second, we will explore how barcode can be used for asset tracking of college devices, equipment, and furniture.

3.1 Student Attendance

Currently, two manual methods are used for taking the student attendance and record the information of students. In the first method, the instructor calls student's name by name during class and record the attendance. Later, after class, the instructor enters the recorded data to university Edugate. Edugate is a portal provided by the university which offers teaching and learning resources to instructors and students. It works as a database for the university administration who keeps students' data which can be later used for retrieval and management purposes. The second method, the instructor, login to Edugate during class and start calling the student's names and record absent students by clicking on the designated box, then the instructor submits the data and logout. Both methods are time-consuming especially when you have a class of 60 students or more. However, the first method takes more time comparing to the second method. The second method requires a good internet connection during class, otherwise the student attendance may take more time or cannot be completed. In this case, the instructor uses the first method and should have a student list with him as a backup to complete the student attendance process.

Now we will discuss the conducted experiment for taking the student attendance with the help of barcode technology. After installing the required applications and software packages (Zint Barcode, Barcode Active program, and Excel), the smartphone and the laptop can be used to conduct the experiment. We took one of the regular classrooms in the college of engineering as an example to conduct the student attendance experiment. The experiment was conducted without the need to generating a barcode because the student ID already has a barcode label. Figure 2 shows the steps of taking student attendance while they are entering the classroom. We used the smartphone, which was linked to our laptop via Wi-Fi, each. Each student must scan the embedded barcode in student ID through the smartphone in a few seconds and get seated.

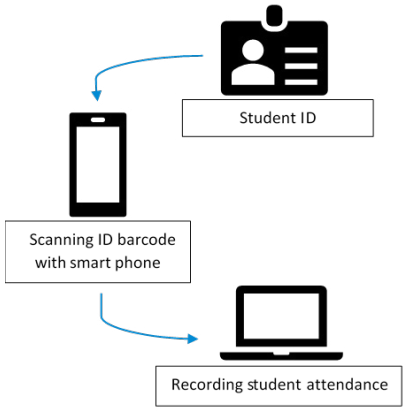

Fig. 2. The steps of taking student attendance.

The barcode-based attendance system is a robust system against cheating intended by the students because the barcode reader is installed closed to the instructor. So, it will be hard for a student to scan ID of his classmates who

are absent. All data during the process of taking the attendance are recorded and stored in the attendance excel spreadsheet and may be linked to Edugate later. During this research, we proposed linking the collected student attendance data to Edugate, but it is not part of this research work.

3. 2 Asset Tracking

To conduct the experiment on a small scale, we choose the College of Engineering meetings room as an example to perform the fixed assets year-end procedures. The process is known as a management system to track college devices, equipment, and furniture and record information. The asset tracking experiment was conducted in two stages, the first stage requires generating, printing, and placing barcode labels on the assets inside the meetings room. The second stage requires reading and recording the fixed assets during the tracking process. Stage one, each barcode generated identify one item (one device or a piece of furniture), then the printed barcode labels are attached to all assets in the meetings room for tracking purposes.

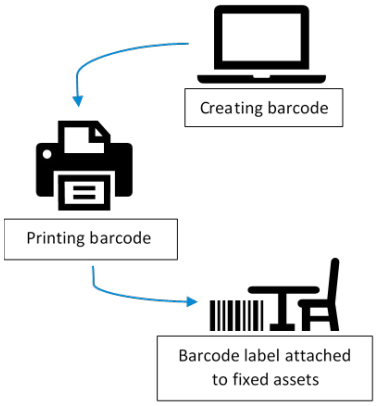

Fig. 3. The sequence of creating and attaching the barcode label on assets.

Figure 3 shows the sequence of generating barcode labels for fixed assets. After generating all barcodes for all assets in the meetings room, the second step is to print them out by barcode printer. Finally, attach the labels on the assets, whether a piece of furniture or a device. Stage two, reading and recording the scanned items of all fixed assets at the college meetings room. The created program for tracking and monitoring provides additional detail about the existing quantity of scanned items and the missing items. Other important information can be recorded during the process, such as the item condition, useful life, and required maintenance.

Table 2. Illustrates the College meetings room fixed assets and its quantity.

Item	Description	Quantity
Swivel Chair	Green	19
Rectangle table	Brown	6
Corner table	Brown	4
Square table	Brown	2
Projector	BENQ	1
Smart board		1
Remote control		1
Laptop table		1
Total		35

Table 2 represents the fixed assets that have been scanned during the experiment and the quantity of each item in the College meetings room. Figure 4 shows the sequence of reading and recording barcode labels for fixed assets through the created system and database.

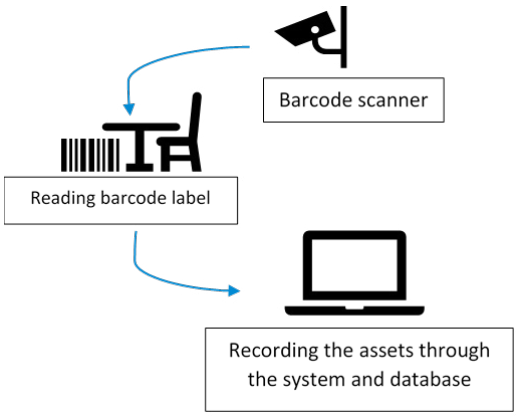

Fig. 4. The sequence of reading and recording assets barcode labels.

3. 3 Assessment of student attendance and asset tracking

Two surveys consisted of questions for measuring faculty members and staff responses for usability, comfort level, and efficiency of the proposed barcode system designed to track student attendance and assets. The study participants consisted of two groups, ten faculty members to measure their response to student attendance experience and ten faculty staff to measure their response to asset tracking experience. All members were male, and the participant's ages ranged from 25 to 55 years old. A research team member provided the created barcode system for faculty members and staff to give it a try along with the related survey to answer the questions in the study. Each survey has questions rating the usability, comfort level, and efficiency on a Likert scale from 1-10. 1) First, we were looking to measure the barcode system's usability for student attendance and asset tracking by faculty and staff, respectively, compared to the manual method. Second, measuring the faculty members and staff comfort level when the barcode system is used for student attendance and asset tracking respectively comparing to the manual method. Third, measuring the efficiency of barcode system when used for student attendance and asset tracking by faculty members and staff respectively (i.e., more or less time, more or less human error, correct or wrong inventory estimates) comparing to the manual method. The results show that the ease of use of barcode system was preferred over the manual method. Participants show a greater comfort level when the barcode system is used over the manual method. Results demonstrated participants preferred barcode system efficiency over the manual method.

4. Conclusion and future work

The implemented barcode system offers a convenient and cost-effective method of tracking students' attendance and assets in a university setting compared to other technologies. The designed barcode system provides a simple methodology to track student attendance and assets. It has several benefits over the manual methods including the process takes less time and stores the data in electronic mode instead of paperwork. It is considered one of the AIDC technologies which provides a high degree of automation, eliminates data entry error, and improve the efficiency of enring data. If the university administration decides to adopt the barcode system proposed in this paper, the system can be integrated with Edugate system of the university to automatically capture and update attendance and assets tracking data. The Edugate database can be used as a resource to provides information to instructors, students, and the university administration.

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